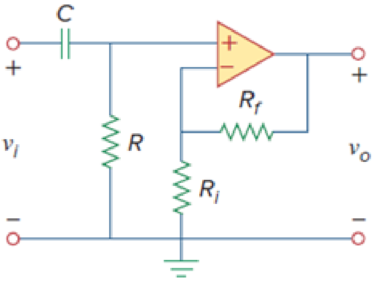
# Chapter 14, Problem 65P

## Problem

A high-pass filter is shown in Fig. 14.92. Show that the transfer function is

$$\mathbf{H}(\omega) = \left(1 + \frac{R_f}{R_i}\right)\frac{j\omega RC}{1 + j\omega RC}$$

**Figure 14.92:**




## Step-by-step solution

**Step 1** of 1

$$V_+ = \frac{R}{R + \frac{1}{j\omega C}}$$

$$V_i = \frac{j\omega RC}{1 + j\omega RC} V_i$$

$$V_- = \frac{R_i}{R_i + R_f} V_o$$

*Since*

$$V_+ = V_-$$

$$\frac{R_i}{R_i + R_f} V_o = \frac{j\omega RC}{1 + j\omega RC} V_i$$

$$H(\omega) = \frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_i}\right)\left(\frac{j\omega RC}{1 + j\omega RC}\right)$$

It is evident that as $\omega \to \infty$

The gain is $\left(1 + \frac{R_f}{R_L}\right)$ and corner frequency is $\frac{1}{\mathrm{RC}}$